# Lecture 10: LTI Resistive Circuits and Equivalent One-Ports

## EE201 – Circuit Theory I

M. Mert Ankaralı

Department of Electrical and Electronics Engineering
Middle East Technical University

current voltage a part of a circuit (box $N$)

# Review: equations from the graph alone

| | **incidence / mesh** | **fundamental cutset / loop** |
|---|---|---|
| KCL | $A\mathbf{i} = \mathbf{0}$ | $Q\mathbf{i} = \mathbf{0}$ |
| KVL | $M\mathbf{v} = \mathbf{0}$ | $B\mathbf{v} = \mathbf{0}$ |
| $n-1$ voltages | node voltages $\mathbf{e}$ | tree voltages $\mathbf{v}_T$ |
| $b-n+1$ currents | mesh currents $\mathbf{i}_M$ | co-tree currents $\mathbf{i}_C$ |
| works for | $M$: planar, nonseparable | any connected graph |

KCL and KVL depend only on the graph: they hold for any elements.

A circuit has $2b$ unknowns: $b$ more equations come from the elements.

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  - substitution theorem, equivalent one-ports
  - series and parallel connections, source transformation

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- **This lecture — transform:** replace a part by a source, or by a simpler part with the same terminal behaviour
  - substitution theorem, equivalent one-ports
  - series and parallel connections, source transformation
- **Later — the circuit as it is:** node voltage and mesh current analysis

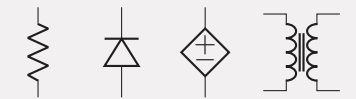
# Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain
2. **A first example** — simplify a circuit step by step
3. **Substitution theorem** — replace a part of a circuit by a source
4. **One-ports and equivalence** — the same $i$–$v$ characteristic at the port
5. **Series and parallel** — combine resistors and sources
6. **One source and one resistor** — exchange a voltage source and a current source

# Resistive and dynamic elements

## Resistive element: no memory

Its law relates $v$ and $i$ at the same instant, algebraically.

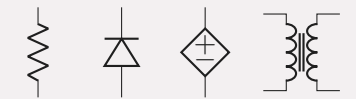



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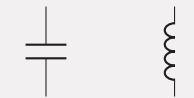
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## Dynamic element: memory

Its law has a derivative.

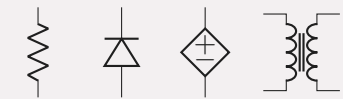



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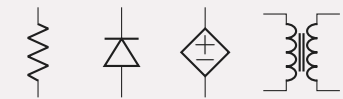

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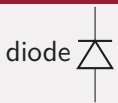

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"Resistive" means memoryless, not only "a resistor".

An element is linear if its law is linear in $v$, $i$; time-invariant if its law does not change with time.

# Three questions about a circuit

## Linear?

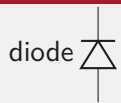



Nonlinear if it contains at least one nonlinear element, e.g. a diode.

Otherwise **linear**.

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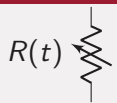
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## Time-invariant?

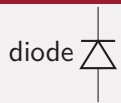



Time-varying if it contains at least one time-varying element, e.g. $R(t)$.

Otherwise **time-invariant**.

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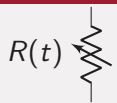
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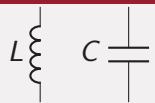
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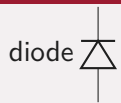



Dynamic if it contains at least one dynamic element: an $L$ or a $C$.

Otherwise **resistive**.

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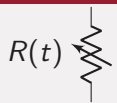
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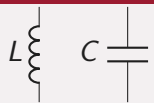
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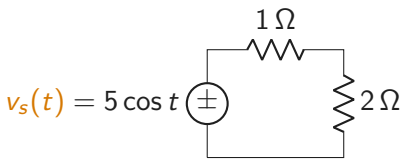

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**LTI resistive circuit = linear + time-invariant + resistive**:

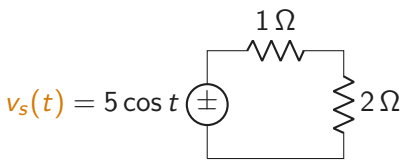
apart from the independent sources, every element is linear, time-invariant and resistive.

# Independent sources are inputs




The source varies in time. Is the circuit time-varying?

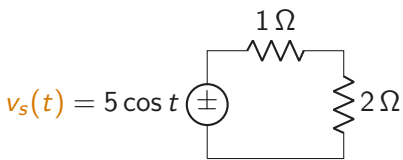
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The three questions are about the circuit that the inputs drive, not about the inputs. This is an LTI resistive circuit.

## What varies in time?

In every circuit $v(t)$ and $i(t)$ are functions of time. The question is whether the element law changes with time.

**Time-invariant resistor**

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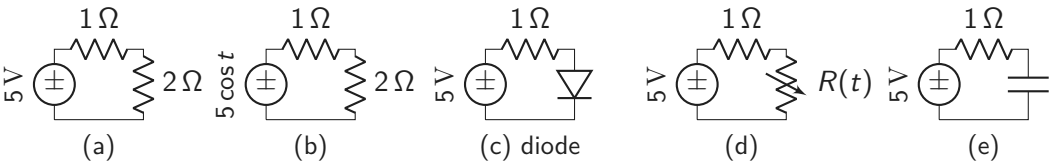
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Linear or nonlinear is also a property of the element law: $v = Ri$ is linear, a diode law is not.

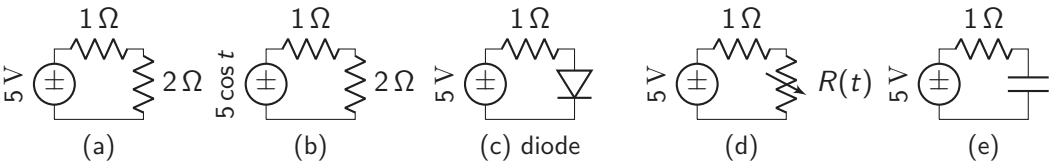
# Which circuit is which?




A source, a $1\,\Omega$ resistor and one more element. In (d), $R(t) = 2 + \sin t\ \Omega$.

Linear? Time-invariant? Resistive?

# Which circuit is which?




| | (a) | (b) | (c) | (d) | (e) |
|---|---|---|---|---|---|
| linear | ✓ | ✓ | × | ✓ | ✓ |
| time-invariant | ✓ | ✓ | ✓ | × | ✓ |
| resistive | ✓ | ✓ | ✓ | ✓ | × |
| LTI resistive | ✓ | ✓ | × | × | × |

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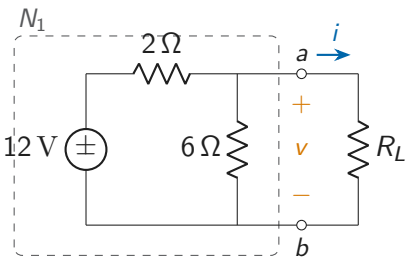
$$v = Ri \ \text{(resistor)}, \qquad v = \mu v_x, \quad i = g v_x, \ \ldots \ \text{(dependent sources)}$$

- Time enters only through the values of the independent sources.
- So at each instant the circuit is solved like a circuit with constant sources: $2b$ linear algebraic equations.

# Roadmap

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2. **A first example** — simplify a circuit step by step
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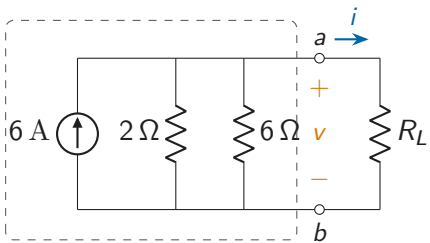
# A first example: simplify the circuit




$R_L$ (the load) can be any resistive element. Find $v$ and $i$.

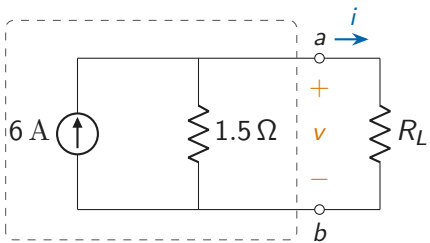
Idea: simplify the part left of $a$–$b$ once.

# A first example: simplify the circuit




**Step 1.** $12\,\mathrm{V}$ in series with $2\,\Omega \rightarrow 6\,\mathrm{A}$ in parallel with $2\,\Omega$ (shown later).

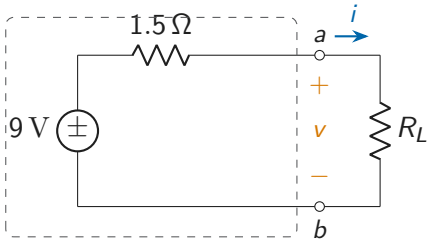
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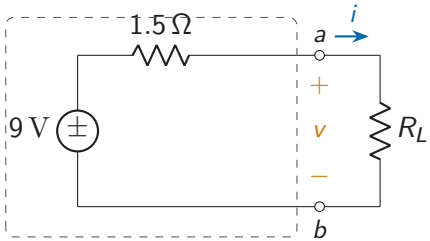



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**Step 3.** $6\,\mathrm{A}$ in parallel with $1.5\,\Omega \rightarrow 9\,\mathrm{V}$ in series with $1.5\,\Omega$.

**Result.** For any $R_L$: $i = \dfrac{9}{1.5 + R_L}$,

$v = R_L i$.

$R_L = 3\,\Omega$: $i = 2\,\mathrm{A}$, $v = 6\,\mathrm{V}$.

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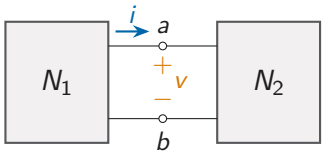
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| question | answered by |
|---|---|
| May a part be replaced by a source? | substitution theorem |
| When may a part be replaced by a simpler part? | equivalent one-ports |
| Which simple replacements are there? | series, parallel, source transformation |

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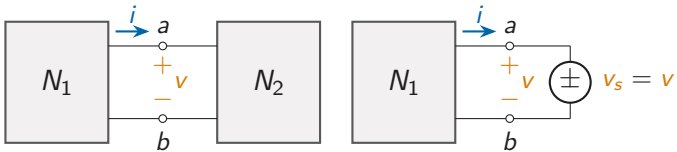
# The substitution theorem




(I)

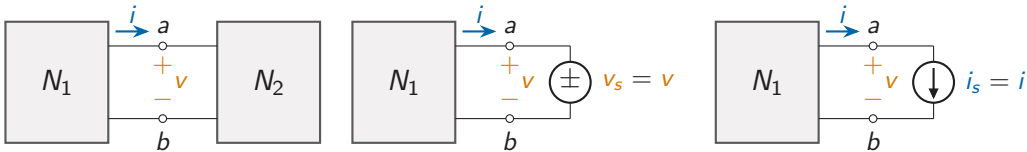
A circuit cut into two parts, joined at the terminals $a$, $b$.
$i$: current from $N_1$ into $N_2$ $v$: voltage between $a$ and $b$

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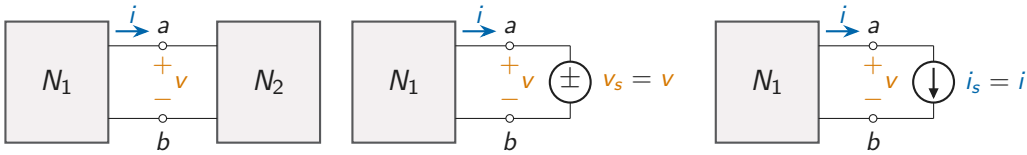

Replace $N_2$ by a voltage source with $v_s(t) = v(t)$ ...

# The substitution theorem




... or by a current source with $i_s(t) = i(t)$.

# The substitution theorem




## Substitution theorem

No branch voltage or branch current inside $N_1$ changes. An observer inside $N_1$ cannot tell (I), (II) and (III) apart.

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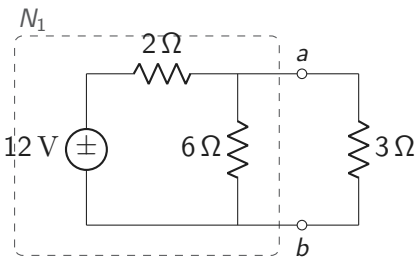
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### One condition

The new circuit must have a unique solution.

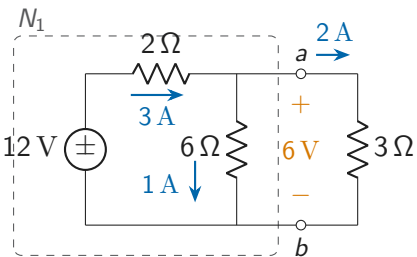
Example: $N_1$ a single current source and $N_2$ replaced by a current source $\Rightarrow$ $v$ undetermined.

# Example: substitution



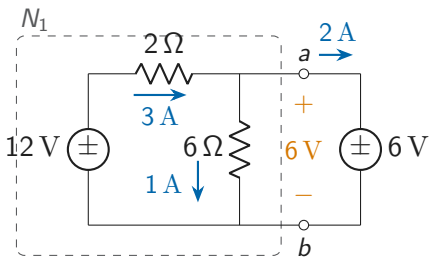

The first example with $R_L = 3\,\Omega$, cut at $a$, $b$: $N_2$ is the load.

# Example: substitution




**(I)** From the first example ($R_L = 3\,\Omega$): $v = 6\,\mathrm{V}$, $i = 2\,\mathrm{A}$; inside $N_1$: $3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$.

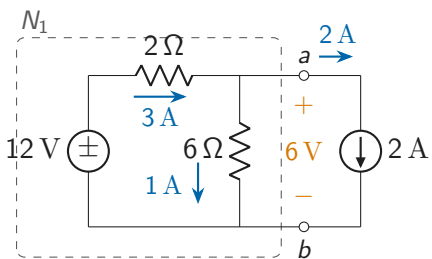
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**(III)** A $2\,\mathrm{A}$ source: KCL at $a$, $\frac{12-v}{2} = \frac{v}{6} + 2$, so $v = 6\,\mathrm{V}$: again $3\,\mathrm{A}$ and $1\,\mathrm{A}$. ✓

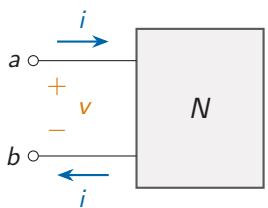
# Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain ✓
2. **A first example** — simplify a circuit step by step ✓
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# Ports and one-ports

## Port, one-port

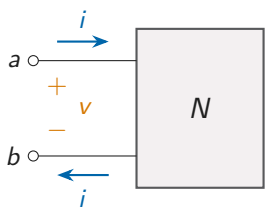
A pair of terminals is a port if the current entering one terminal equals the current leaving the other. A one-port is a circuit, or a circuit element, with one pair of terminals accessible from outside.



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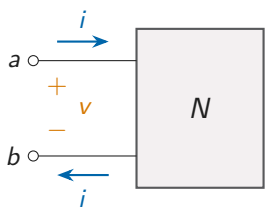



- A one-port is always a port: KCL on a closed surface around $N$.

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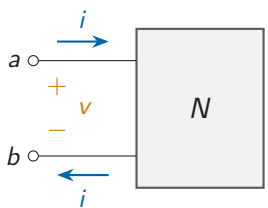



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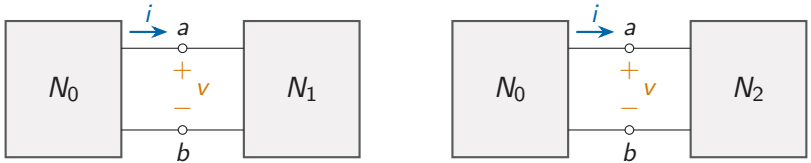



- A one-port is always a port: KCL on a closed surface around $N$.
- $i$ enters the $+$ terminal of $v$.
- The relation between $v$ and $i$ is the $i$–$v$ characteristic, e.g. $v = Ri$ for a resistor.

# Equivalent one-ports

## Equivalent one-ports

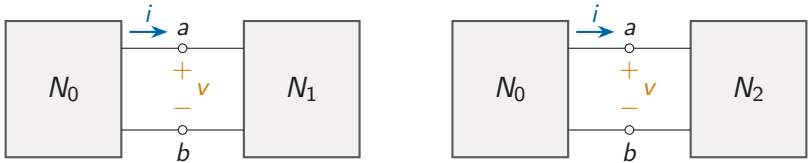
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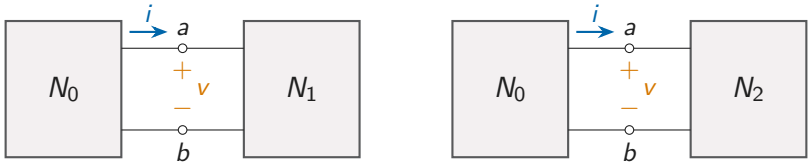



For any $N_0$: every branch voltage and current inside $N_0$ is the same in both.

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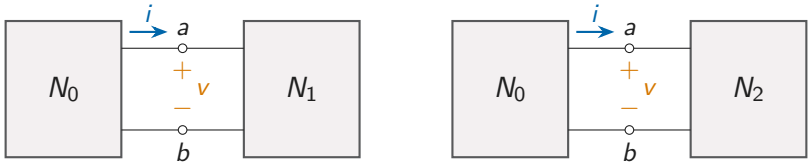

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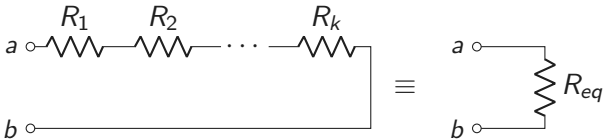
Inside $N_1$ and $N_2$ the branch values may differ.

# Roadmap

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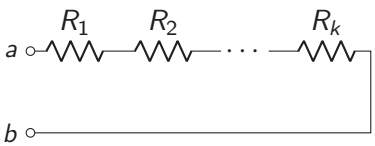
# Series connection

Same current $i$ through every element (KCL); the voltages add (KVL).

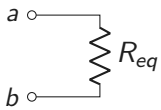


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$\equiv$

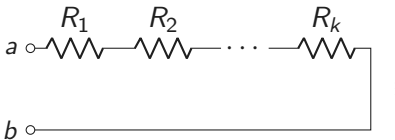



$$R_{eq} = R_1 + R_2 + \cdots + R_k$$

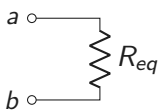
$$v = R_1 i + R_2 i + \cdots + R_k i = R_{eq} i$$

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$\equiv$

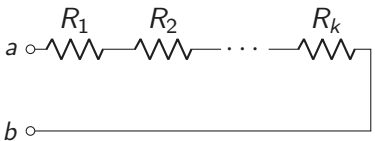



$$R_{eq} = R_1 + R_2 + \cdots + R_k$$

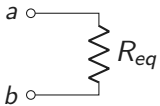
Voltage divider: the share of $R_j$ is $v_j = \dfrac{R_j}{R_{eq}}\, v$

# Series connection

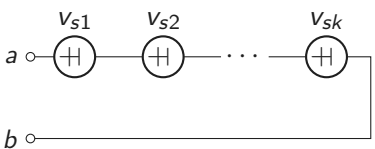
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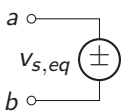

$\equiv$




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$\equiv$

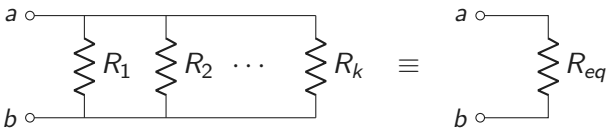



$$v_{s,eq} = v_{s1} + v_{s2} + \cdots + v_{sk}$$

All $+$ signs towards $a$; a source turned the other way enters with $-$.

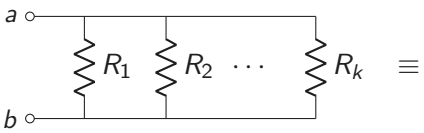
# Parallel connection

Same voltage $v$ across every element (KVL); the currents add (KCL).



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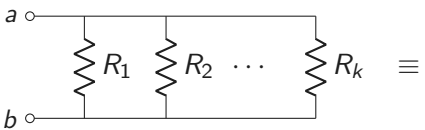



$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_k}$$

With $G_j = 1/R_j$: $\quad i = G_1 v + \cdots + G_k v = G_{eq} v, \quad G_{eq} = G_1 + \cdots + G_k$

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Same voltage $v$ across every element (KVL); the currents add (KCL).

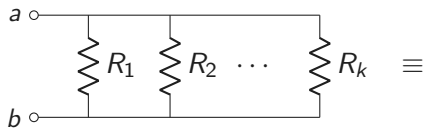



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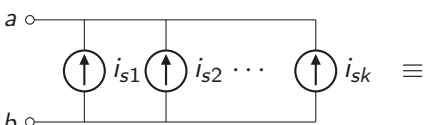
Current divider: $i_j = \dfrac{G_j}{G_{eq}}\, i.$ Two resistors: $R_{eq} = \dfrac{R_1 R_2}{R_1 + R_2}$

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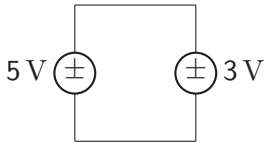

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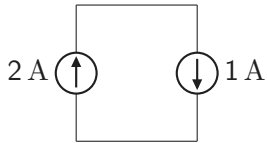

$$i_{s,eq} = i_{s1} + i_{s2} + \cdots + i_{sk}$$

All arrows towards $a$; a source turned the other way enters with $-$.

# Two connections to avoid



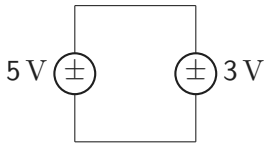

two voltage sources in parallel




two current sources in series

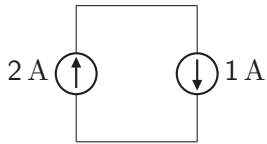
Are these allowed?

# Two connections to avoid




two voltage sources in parallel

KVL fails: $5 \neq 3$.




two current sources in series

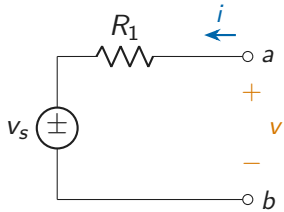
KCL fails: $2 \neq 1$.

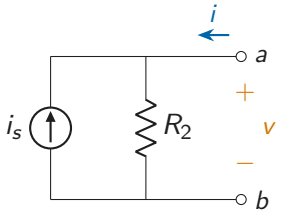
With equal values they are allowed, but the shared current (or voltage) is not fixed.

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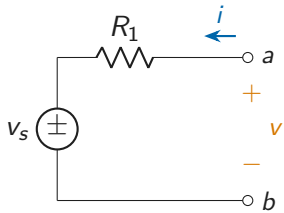
# One source and one resistor

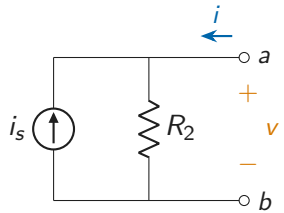



When do these two one-ports have the same $i$–$v$ characteristic?

# One source and one resistor

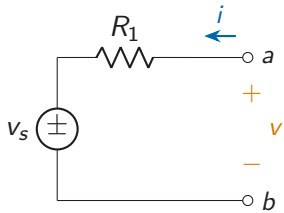


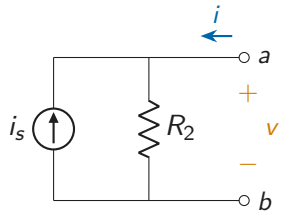

$$v = R_1 i + v_s$$

KVL around the loop: $R_1$ carries $i$.

# One source and one resistor



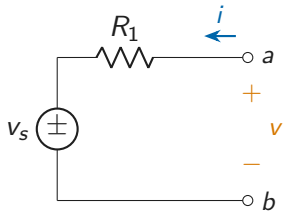

$v = R_1 i + v_s$



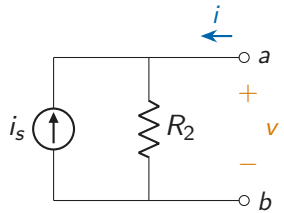

$v = R_2(i + i_s) = R_2 i + R_2 i_s$

$i$ and $i_s$ both flow into the top node: $R_2$ carries $i + i_s$.

# One source and one resistor




$$v = R_1 i + v_s$$




$$v = R_2(i + i_s) = R_2 i + R_2 i_s$$

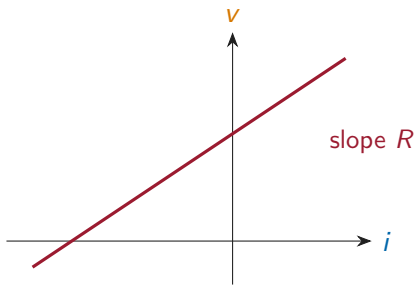
Both are straight lines in the $i$–$v$ plane: equal slopes and equal intercepts.

# Source transformation

## Source transformation

The two one-ports are equivalent if

$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$



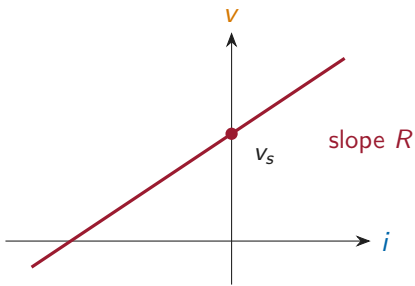
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Open circuit ($i = 0$): $v = v_s$.



# Source transformation

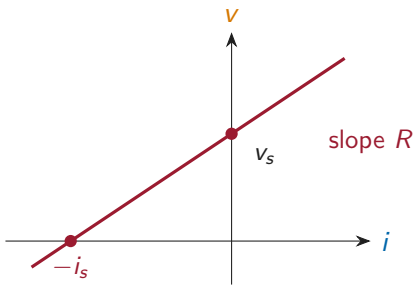
## Source transformation

The two one-ports are equivalent if

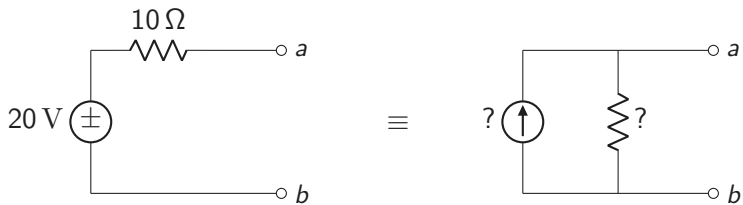
$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$

Open circuit ($i = 0$): $v = v_s$.

Short circuit ($v = 0$): $i = -i_s$.

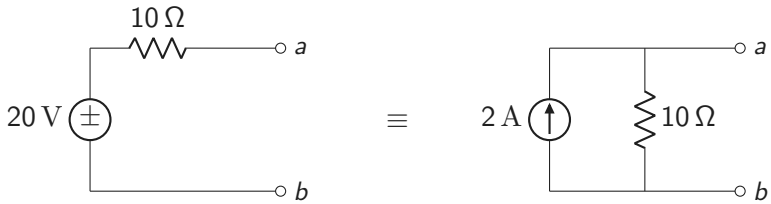


# Example: a source transformation




Find the current source and the resistor.

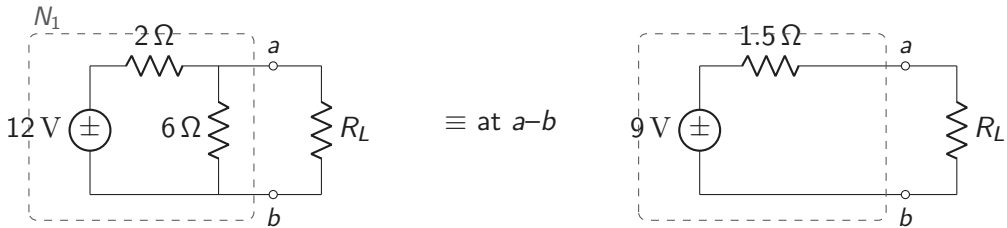
# Example: a source transformation




$$R = 10\,\Omega, \qquad i_s = v_s/R = 20/10 = 2\,\mathrm{A}$$

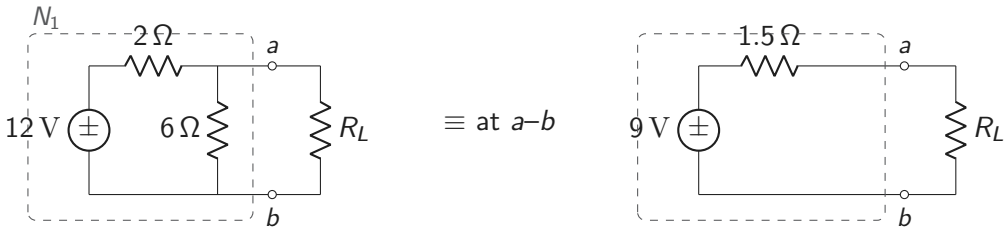
Both: open-circuit voltage $20\,\mathrm{V}$, short-circuit current $2\,\mathrm{A}$ (out of $a$).

# The first example, justified



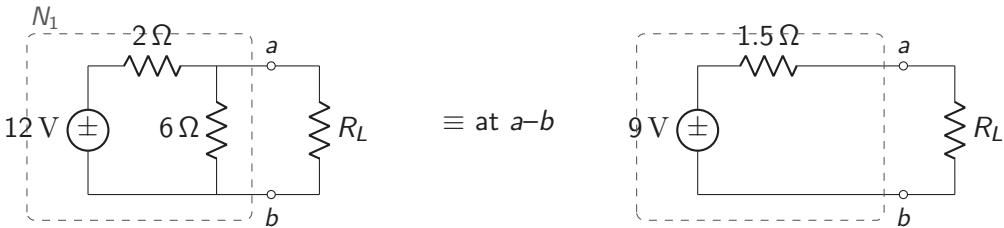

- Steps 1 and 3 are source transformations: $12 = 2 \times 6$ and $9 = 1.5 \times 6$.

# The first example, justified



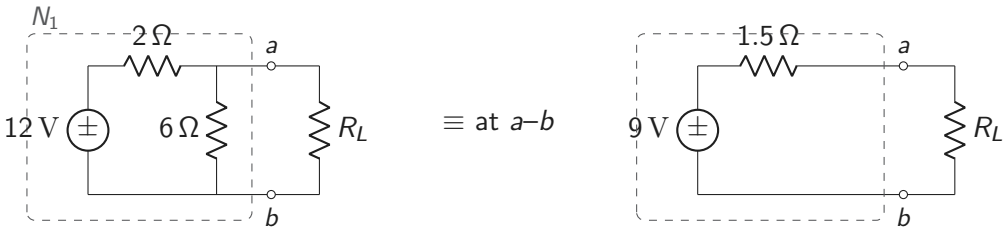

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- Step 2 is a parallel connection: $2\,\Omega \parallel 6\,\Omega = 1.5\,\Omega$.

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- Steps 1 and 3 are source transformations: $12 = 2 \times 6$ and $9 = 1.5 \times 6$.
- Step 2 is a parallel connection: $2\,\Omega \parallel 6\,\Omega = 1.5\,\Omega$.
- So the two parts left of *a*–*b* are equivalent one-ports.
- Careful: inside them the values differ; for $R_L = 3\,\Omega$ the $12\,\mathrm{V}$ source carries $3\,\mathrm{A}$, the $9\,\mathrm{V}$ source $2\,\mathrm{A}$.

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- Both start from the matrices met earlier ($A$, $M$), now with linear element equations.

# Summary

- LTI resistive: apart from the independent sources (external inputs), every element is linear, time-invariant and memoryless, so all equations are linear and algebraic.
- Substitution: replace $N_2$ by a voltage source $v_s = v$ or a current source $i_s = i$; nothing changes inside $N_1$.
- Equivalent one-ports: the same $i$–$v$ characteristic; the rest cannot tell them apart.
- Series: $R_{eq} = \sum R_j$, $v_{s,eq} = \sum v_{sj}$. Parallel: $G_{eq} = \sum G_j$, $i_{s,eq} = \sum i_{sj}$.
- Source transformation: $v_s$ with series $R \equiv i_s = v_s/R$ with parallel $R$.